Elasticity of floppy networks of rigid bars

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1 Definitions

(Side note: I will write down what we discussed in the first two meetings. Please let me know if I misunderstand something.)

1.1 General Description

$z < z_c = 2đ$

The (linearized) rigid-bar contraint ensures:

$(\vec{u}_j - \vec{u}_i) \cdot \hat{n}_{ij} = 0$

Derivation: $0 = \delta r_{ij} = \frac{\partial r_{ij}}{\partial x_k} \cdot \vec{u}_k \left[+ \frac{\partial^2 r_{ij}}{\partial x_k \partial x_l} \cdot \vec{u}_k \vec{u}_l + ... \right]$

We want to eventually obtain "nodal accelerations" $|A\rangle$ (i.e., it lives in node space)

For this we need to make an assumption for the "nodal velocities" $|V\rangle$ such that it account for the compatibiltiy matrix S,

$$\frac{\partial r_{ij}}{\partial x_k} = S_{ij,k} = S_{\alpha,k} = (\delta_{jk} - \delta_{ik}) \frac{x_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik}) \hat{n}_{ij}.$$

Thus, we require:

$$S|V\rangle = 0$$

1.1.1 How to obtain the nodal velocities?

Consider bar space and node space eigenvectors respectively (of the matrices SS^T and S^TS):

$$SS^T|\varphi_m\rangle = \omega_m^2 |\varphi_m\rangle \tag{1}$$

$$S^TS|\Psi_m\rangle = \omega_m^2 |\Psi_m\rangle \tag{2}$$

Note (prove later, it is trivial): SS^T and S^TS are symmetric and positive semi-definite Proof: (1) Symmetric, because $(SS^T)^T = (S^T)^T S^T = SS^T$ and $(S^TS)^T = S^T(S^T)^T = S^TS$. (2) Positive semi-definite, because SS^T and S^TS are $N_C \times N_C$ and $đN \times đN$ symmetric matrices respectively. For a symmetric $n \times n$ matrix A, we have $x^TAx = x^T(PDP^{-1})x = y^TDy = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 > 0$ after setting $y = P^Tx$ (principal axes theorem)

We can derive the following two identities:

$$S|\Psi_m\rangle = \omega_m |\varphi_m\rangle \tag{3}$$

Proof: $S(S^TS|\Psi_m\rangle) = S(\omega_m^2|\Psi_m\rangle) \Rightarrow (SS^T)S|\Psi_m\rangle = \omega_m^2 S|\Psi_m\rangle$. This means that $S|\Psi_m\rangle$ is an eigenvector in node space and we can say $S|\Psi_m\rangle = c|\varphi_m\rangle$. We can calculate $||S|\Psi_m\rangle||^2 = \langle\Psi_m|S^TS|\Psi_m\rangle = \omega^2 = c^2\langle\varphi_m|\varphi_m\rangle \Rightarrow c = \omega_m$.

$$S^T|\varphi_m\rangle = \omega_m |\Psi_m\rangle \tag{4}$$

Proof: Similar to first identity

Thus, SS^T and S^TS have the same nonzero eigenvalues

Proof: By virtue of the identities in equations 3 and 4

We can also see this by knowing the number of eigenvectors with a non-zero eigenvalue is equal to the rank of the matrix and that

$\text{rank}(S) = \text{rank}(S^T S) = \text{rank}(SS^T)$

Furthermore, we have

$$S \in \mathbb{R}^{N_C \times đN}, SS^T \in \mathbb{R}^{N_C \times N_C}, S^T S \in \mathbb{R}^{đN \times đN}$$

with N_C the number of constraints/bars, $đ$ dimensionality of the system and N the number of nodes/particles.

S can be see as a linear operator mapping vectors from node space to bar space:

$$S|V\rangle = |v\rangle$$

S is square when $2N_C = đN \Rightarrow z = \frac{2N_C}{N} = 2đ \equiv z_c$

States of self-stress span the kernel of S^T (and nullity(S^T) = number of SSS's)

Floppy modes span the kernel of S (and nullity(S) = number of floppy modes).

Now, we know that SS^T is symmetric and because of Eq. 1 we know that it has an eigenbasis $\{|\varphi_m\rangle\}$. So, we can write the following singular value decomposition (SVD) in braket notation (remembering that here $z < z_c$):

$$(SS^T)^{-1} = \sum_{\omega_m > 0} \frac{|\varphi_m\rangle\langle\varphi_m|}{\omega_m^2} \tag{5}$$

Now we write

$$S^T(SS^T)^{-1}S = \sum_{\omega_m>0} \frac{S|\varphi_m\rangle\langle\varphi_m|S^T}{\omega_m^2} = \sum_{\omega_m>0} \frac{\omega_m^2|\Psi_m\rangle\langle\Psi_m|}{\omega_m^2} = Q$$

$Q = \sum_{\omega_m>0} |\Psi_m\rangle\langle\Psi_m|$ is an operator that is idempotent (i.e., $Q^2 = Q$)

Note: the range of Q is a subspace of $\mathbb{R}^{đN}$ (corresponding to the range of S^T or row space of S. It basically removes floppy modes and only keeps motions affecting constraints.

We can then define the complement projection of Q, $I - Q = P = I - S^T(SS^T)^{-1}S$ which will project onto the nullspace of S, Ker(S) or Nul(S). P will essentially remove all motions affecting bar lengths and keep floppy modes. We have $P = \sum_{\omega_m=0} |\Psi_m\rangle\langle\Psi_m|$ and it will project onto space spanned by zero modes of $S^T S$, $\ker(P) = \ker(S)$.

[Any mode $|u\rangle$ can be decomposed as $Q|u\rangle + P|u\rangle$]

To obtain $|V\rangle$ s.t. $S|V\rangle = 0$

Following steps:

$$S|V\rangle = S(P|V'\rangle) = S(I - S^T(SS^T)^{-1}S)|V'\rangle$$

(

$$\Rightarrow S|V'\rangle - SS^T(SS^T)^{-1}S|V'\rangle = 0$$

, indeed satisfies homogeneous equation)

1) Initialize $|V'\rangle$ Compute $|V_1\rangle = S|V'\rangle$

2) Compute $|V_2\rangle$ s.t. $SS^T|V_2\rangle = |V_1\rangle \Rightarrow |V_2\rangle = (SS^T)^{-1}|V_1\rangle$

3) Obtain deisred $|V\rangle$: $|V\rangle = |V'\rangle - S^T|V_2\rangle$

4) Ensure $S|V\rangle = 0$

1.2 Continuation: nodal accelerations and shear modulus

$$\frac{d}{dt}(S|V\rangle) = \frac{dS}{dt}|V\rangle + S|A\rangle = 0$$

$$\frac{dS}{dt} = \frac{\partial S}{\partial x}|V\rangle$$

(Time-dependence of S arises from motion of the nodes)

$$\left[\langle V|\frac{\partial S}{\partial x}|V\rangle\right]_\alpha + S|A\rangle = 0$$

$$\vec{A}_j = \sum_{i \to j} f_{ij} \hat{n}_i$$

$$\Rightarrow |A\rangle = S^T |f\rangle = -S^T (SS^T)^{-1} \left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha$$

(Dynamical equation)

$$\left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha + SS^T |f\rangle = 0$$

$$\Rightarrow |f\rangle = -(SS^T)^{-1} \left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha$$

$$|A\rangle = -S^T (SS^T)^{-1} \left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha = \langle V | O | V \rangle$$

$$O^A_{ijk} = -S^T_\alpha (SS^T)^{-1}_{\alpha\beta} \frac{\partial S_{\beta,j}}{\partial x_k}$$

(effective operator that maps velocities to accelerations under the rigidity constraints.)

$$\Gamma = \frac{\langle \frac{\partial r}{\partial \gamma} | f \rangle}{Volume}$$

with γ the strain and Γ as stress (stress computed from internal forces due to an imposed strain γ)

the shear modulus:

$$G = \frac{\Gamma_2 - \Gamma_1}{\delta \gamma}$$

We want to find $G(z)$

2 How to compute (nodal) forces and accelerations?

Third rank tensor:

$$\frac{\partial S_{ij,k}}{\partial x_l} = (\delta_{jk} - \delta_{ik}) \partial_{x_k} (\frac{x_{ij}}{r_{ij}})$$

$$\frac{\partial S_{ij,k}}{\partial \vec{x}_l} = (\delta_{jk} - \delta_{ik})(\delta_{jl} - \delta_{il}) \cdot \frac{1}{r_{ij}} \left(I - \hat{n}_{ij} \hat{n}^T_{ij} \right)$$

Now this requires a double contraction and this for a specific $\alpha \in \{1, \ldots, N_c\}$

$$\left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha = \sum_{k,l} V_k \frac{\partial S_{\alpha,k}}{\partial x_l} V_l = \sum_{k,l} V_k \frac{\partial S_{ij,k}}{\partial x_l} V_l = \sum_{k,l} (u_j - u_i)^T (u_j - u_i) \cdot \frac{1}{r_{ij}} \left(I - \hat{n}_{ij} \hat{n}^T_{ij} \right)$$

$$= \sum_{k,l} \frac{1}{r_{ij}} (u_{ij} \cdot u_{ij} - (u_{ij} \cdot \hat{n}_{ij})^2)$$

and then we need to simply compute

$$|f\rangle = -(SS^T)^{-1} \left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha$$

and then $|A\rangle = S^T |f\rangle = -S^T (SS^T)^{-1} \left[\langle V | \frac{\partial S}{\partial x} | V \rangle \right]_\alpha$

Notice that: $|f\rangle \in \mathbb{R}^{N_c}$ and $|A\rangle \in \mathbb{R}^{đN}$

Then, for the shear stress Γ we need to evaluate $\langle \frac{\partial r}{\partial \gamma} |$